

Integration of Rational Functions by Partial Fractions

In this section we show how to integrate any rational function (a ratio of polynomials) by expressing it as a sum of simpler fractions, called *partial fractions*, that we already know how to integrate. To illustrate the method, observe that by taking the fractions $2/(x - 1)$ and $1/(x + 2)$ to a common denominator we obtain

$$\frac{2}{x - 1} - \frac{1}{x + 2} = \frac{2(x + 2) - (x - 1)}{(x - 1)(x + 2)} = \frac{x + 5}{x^2 + x - 2}$$

If we now reverse the procedure, we see how to integrate the function on the right side of this equation:

$$\begin{aligned}\int \frac{x + 5}{x^2 + x - 2} dx &= \int \left(\frac{2}{x - 1} - \frac{1}{x + 2} \right) dx \\ &= 2 \ln |x - 1| - \ln |x + 2| + C\end{aligned}$$

To see how the method of partial fractions works in general, let's consider a rational function

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials. It's possible to express f as a sum of simpler fractions provided that the degree of P is less than the degree of Q . Such a rational function is called *proper*. Recall that if

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $a_n \neq 0$, then the degree of P is n and we write $\deg(P) = n$.

If f is improper, that is, $\deg(P) \geq \deg(Q)$, then we must take the preliminary step of dividing P into Q (by long division) until a remainder $R(x)$ is obtained such that $\deg(R) < \deg(Q)$. The division statement is

$$\boxed{1} \qquad f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

where S and R are also polynomials.

EXAMPLE 1 Find $\int \frac{x^3 + x}{x - 1} dx$.

SOLUTION Since the degree of the numerator is greater than the degree of the denominator, we first perform the long division. This enables us to write

$$\begin{aligned}\int \frac{x^3 + x}{x - 1} dx &= \int \left(x^2 + x + 2 + \frac{2}{x - 1} \right) dx \\ &= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln |x - 1| + C\end{aligned}$$

The next step is to factor the denominator $Q(x)$ as far as possible. It can be shown that any polynomial Q can be factored as a product of linear factors (of the form $ax + b$) and irreducible quadratic factors (of the form $ax^2 + bx + c$, where $b^2 - 4ac < 0$). For instance, if $Q(x) = x^4 - 16$, we could factor it as

$$Q(x) = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$$

The third step is to express the proper rational function $R(x)/Q(x)$ (from Equation 1) as a sum of **partial fractions** of the form

$$\frac{A}{(ax + b)^i} \quad \text{or} \quad \frac{Ax + B}{(ax^2 + bx + c)^j}$$

A theorem in algebra guarantees that it is always possible to do this. We explain the details for the four cases that occur.

CASE 1 • The denominator $Q(x)$ is a product of distinct linear factors.

This means that we can write

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_kx + b_k)$$

where no factor is repeated (and no factor is a constant multiple of another). In this case the partial fraction theorem states that there exist constants $A_1, A_2, \dots, A_k$ such that

$$\boxed{2} \quad \frac{R(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \cdots + \frac{A_k}{a_kx + b_k}$$

These constants can be determined as in the following example.

EXAMPLE 2 Evaluate $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$.

SOLUTION Since the degree of the numerator is less than the degree of the denominator, we don't need to divide. We factor the denominator as

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(2x - 1)(x + 2)$$

Since the denominator has three distinct linear factors, the partial fraction decomposition of the integrand (2) has the form

$$\boxed{3} \quad \frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

To determine the values of A , B , and C , we multiply both sides of this equation by the product of the denominators, $x(2x - 1)(x + 2)$, obtaining

$$\boxed{4} \quad x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

Expanding the right side of Equation 4 and writing it in the standard form for polynomials, we get

$$\boxed{5} \quad x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A$$

The polynomials in Equation 5 are identical, so their coefficients must be equal. The coefficient of x^2 on the right side, $2A + B + 2C$, must equal the coefficient of x^2 on the left side—namely, 1. Likewise, the coefficients of x are equal and the constant terms are equal. This gives the following system of equations for A , B , and C :

$$2A + B + 2C = 1$$

$$3A + 2B - C = 2$$

$$-2A = -1$$

Solving, we get $A = \frac{1}{2}$, $B = \frac{1}{5}$, and $C = -\frac{1}{10}$, and so

$$\begin{aligned}\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx &= \int \left(\frac{1}{2} \frac{1}{x} + \frac{1}{5} \frac{1}{2x - 1} - \frac{1}{10} \frac{1}{x + 2} \right) dx \\ &= \frac{1}{2} \ln |x| + \frac{1}{10} \ln |2x - 1| - \frac{1}{10} \ln |x + 2| + K\end{aligned}$$

In integrating the middle term we have made the mental substitution $u = 2x - 1$, which gives $du = 2 dx$ and $dx = du/2$.

EXAMPLE 3 Find $\int \frac{dx}{x^2 - a^2}$, where $a \neq 0$.

SOLUTION The method of partial fractions gives

$$\frac{1}{x^2 - a^2} = \frac{1}{(x - a)(x + a)} = \frac{A}{x - a} + \frac{B}{x + a}$$

and therefore

$$A(x + a) + B(x - a) = 1$$

Using the method of the preceding note, we put $x = a$ in this equation and get $A(2a) = 1$, so $A = 1/(2a)$. If we put $x = -a$, we get $B(-2a) = 1$, so $B = -1/(2a)$. Thus

$$\begin{aligned}\int \frac{dx}{x^2 - a^2} &= \frac{1}{2a} \int \left(\frac{1}{x - a} - \frac{1}{x + a} \right) dx \\ &= \frac{1}{2a} (\ln |x - a| - \ln |x + a|) + C\end{aligned}$$

Since $\ln x - \ln y = \ln(x/y)$, we can write the integral as

$$\boxed{6} \quad \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C$$

CASE II • $Q(x)$ is a product of linear factors, some of which are repeated.

Suppose the first linear factor $(a_1x + b_1)$ is repeated r times; that is, $(a_1x + b_1)^r$ occurs in the factorization of $Q(x)$. Then instead of the single term $A_1/(a_1x + b_1)$ in Equation 2, we would use

$$\boxed{7} \quad \frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}$$

By way of illustration, we could write

$$\frac{x^3 - x + 1}{x^2(x - 1)^3} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x - 1} + \frac{D}{(x - 1)^2} + \frac{E}{(x - 1)^3}$$

but we prefer to work out in detail a simpler example.

EXAMPLE 4 Find $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx$.

SOLUTION The first step is to divide. The result of long division is

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}$$

The second step is to factor the denominator $Q(x) = x^3 - x^2 - x + 1$. Since $Q(1) = 0$, we know that $x - 1$ is a factor and we obtain

$$\begin{aligned} x^3 - x^2 - x + 1 &= (x - 1)(x^2 - 1) = (x - 1)(x - 1)(x + 1) \\ &= (x - 1)^2(x + 1) \end{aligned}$$

Since the linear factor $x - 1$ occurs twice, the partial fraction decomposition is

$$\frac{4x}{(x - 1)^2(x + 1)} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{x + 1}$$

Multiplying by the least common denominator, $(x - 1)^2(x + 1)$, we get

$$\begin{aligned} \boxed{8} \quad 4x &= A(x - 1)(x + 1) + B(x + 1) + C(x - 1)^2 \\ &= (A + C)x^2 + (B - 2C)x + (-A + B + C) \end{aligned}$$

Now we equate coefficients:

$$\begin{aligned} A + C &= 0 \\ B - 2C &= 4 \\ -A + B + C &= 0 \end{aligned}$$

Solving, we obtain $A = 1$, $B = 2$, and $C = -1$, so

$$\begin{aligned}\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx &= \int \left[x + 1 + \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} \right] dx \\ &= \frac{x^2}{2} + x + \ln |x-1| - \frac{2}{x-1} - \ln |x+1| + K \\ &= \frac{x^2}{2} + x - \frac{2}{x-1} + \ln \left| \frac{x-1}{x+1} \right| + K\end{aligned}$$

CASE III • $Q(x)$ contains irreducible quadratic factors, none of which is repeated.

If $Q(x)$ has the factor $ax^2 + bx + c$, where $b^2 - 4ac < 0$, then, in addition to the partial fractions in Equations 2 and 7, the expression for $R(x)/Q(x)$ will have a term of the form

$$\boxed{9} \quad \frac{Ax + B}{ax^2 + bx + c}$$

where A and B are constants to be determined. For instance, the function given by $f(x) = x/[(x - 2)(x^2 + 1)(x^2 + 4)]$ has a partial fraction decomposition of the form

$$\frac{x}{(x - 2)(x^2 + 1)(x^2 + 4)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{x^2 + 4}$$

The term given in (9) can be integrated by completing the square and using the formula

$\boxed{10}$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

EXAMPLE 5 Evaluate $\int \frac{2x^2 - x + 4}{x^3 + 4x} dx$.

SOLUTION Since $x^3 + 4x = x(x^2 + 4)$ can't be factored further, we write

$$\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

Multiplying by $x(x^2 + 4)$, we have

$$\begin{aligned} 2x^2 - x + 4 &= A(x^2 + 4) + (Bx + C)x \\ &= (A + B)x^2 + Cx + 4A \end{aligned}$$

Equating coefficients, we obtain

$$A + B = 2 \qquad C = -1 \qquad 4A = 4$$

Thus $A = 1$, $B = 1$, and $C = -1$ and so

$$\int \frac{2x^2 - x + 4}{x^3 + 4x} dx = \int \left(\frac{1}{x} + \frac{x - 1}{x^2 + 4} \right) dx$$

In order to integrate the second term we split it into two parts:

$$\int \frac{x-1}{x^2+4} dx = \int \frac{x}{x^2+4} dx - \int \frac{1}{x^2+4} dx$$

We make the substitution $u = x^2 + 4$ in the first of these integrals so that $du = 2x dx$.

We evaluate the second integral by means of Formula 10 with $a = 2$:

$$\begin{aligned} \int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx &= \int \frac{1}{x} dx + \int \frac{x}{x^2 + 4} dx - \int \frac{1}{x^2 + 4} dx \\ &= \ln |x| + \frac{1}{2} \ln(x^2 + 4) - \frac{1}{2} \tan^{-1}(x/2) + K \end{aligned}$$

EXAMPLE 6 Evaluate $\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx$.

SOLUTION Since the degree of the numerator is *not less than* the degree of the denominator, we first divide and obtain

$$\frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} = 1 + \frac{x - 1}{4x^2 - 4x + 3}$$

Notice that the quadratic $4x^2 - 4x + 3$ is irreducible because its discriminant is $b^2 - 4ac = -32 < 0$. This means it can't be factored, so we don't need to use the partial fraction technique.

To integrate the given function we complete the square in the denominator:

$$4x^2 - 4x + 3 = (2x - 1)^2 + 2$$

This suggests that we make the substitution $u = 2x - 1$. Then, $du = 2 dx$ and $x = (u + 1)/2$, so

$$\begin{aligned} \int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx &= \int \left(1 + \frac{x - 1}{4x^2 - 4x + 3} \right) dx \\ &= x + \frac{1}{2} \int \frac{\frac{1}{2}(u + 1) - 1}{u^2 + 2} du = x + \frac{1}{4} \int \frac{u - 1}{u^2 + 2} du \\ &= x + \frac{1}{4} \int \frac{u}{u^2 + 2} du - \frac{1}{4} \int \frac{1}{u^2 + 2} du \\ &= x + \frac{1}{8} \ln(u^2 + 2) - \frac{1}{4} \cdot \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{u}{\sqrt{2}} \right) + C \\ &= x + \frac{1}{8} \ln(4x^2 - 4x + 3) - \frac{1}{4\sqrt{2}} \tan^{-1} \left(\frac{2x - 1}{\sqrt{2}} \right) + C \end{aligned}$$

NOTE ◦ Example 6 illustrates the general procedure for integrating a partial fraction of the form

$$\frac{Ax + B}{ax^2 + bx + c} \quad \text{where } b^2 - 4ac < 0$$

We complete the square in the denominator and then make a substitution that brings the integral into the form

$$\int \frac{Cu + D}{u^2 + a^2} du = C \int \frac{u}{u^2 + a^2} du + D \int \frac{1}{u^2 + a^2} du$$

Then the first integral is a logarithm and the second is expressed in terms of $\tan^{-1}$.

CASE IV • $Q(x)$ contains a repeated irreducible quadratic factor.

If $Q(x)$ has the factor $(ax^2 + bx + c)^r$, where $b^2 - 4ac < 0$, then instead of the single partial fraction (9), the sum

$$(11) \quad \frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$$

occurs in the partial fraction decomposition of $R(x)/Q(x)$. Each of the terms in (11) can be integrated by first completing the square.

EXAMPLE 7 Write out the form of the partial fraction decomposition of the function

$$\frac{x^3 + x^2 + 1}{x(x - 1)(x^2 + x + 1)(x^2 + 1)^3}$$

SOLUTION

$$\begin{aligned} & \frac{x^3 + x^2 + 1}{x(x - 1)(x^2 + x + 1)(x^2 + 1)^3} \\ &= \frac{A}{x} + \frac{B}{x - 1} + \frac{Cx + D}{x^2 + x + 1} + \frac{Ex + F}{x^2 + 1} + \frac{Gx + H}{(x^2 + 1)^2} + \frac{Ix + J}{(x^2 + 1)^3} \end{aligned}$$

EXAMPLE 8 Evaluate $\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx$.

SOLUTION The form of the partial fraction decomposition is

$$\frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying by $x(x^2 + 1)^2$, we have

$$\begin{aligned} -x^3 + 2x^2 - x + 1 &= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x \\ &= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex \\ &= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A \end{aligned}$$

If we equate coefficients, we get the system

$$A + B = 0 \qquad C = -1 \qquad 2A + B + D = 2 \qquad C + E = -1 \qquad A = 1$$

which has the solution $A = 1$, $B = -1$, $C = -1$, $D = 1$, and $E = 0$. Thus

$$\begin{aligned}
\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx &= \int \left(\frac{1}{x} - \frac{x + 1}{x^2 + 1} + \frac{x}{(x^2 + 1)^2} \right) dx \\
&= \int \frac{dx}{x} - \int \frac{x}{x^2 + 1} dx - \int \frac{dx}{x^2 + 1} + \int \frac{x dx}{(x^2 + 1)^2} \\
&= \ln |x| - \frac{1}{2} \ln(x^2 + 1) - \tan^{-1} x - \frac{1}{2(x^2 + 1)} + K
\end{aligned}$$

In the second and fourth terms we made the mental substitution $u = x^2 + 1$.

We note that sometimes partial fractions can be avoided when integrating a rational function. For instance, although the integral

$$\int \frac{x^2 + 1}{x(x^2 + 3)} dx$$

could be evaluated by the method of Case III, it's much easier to observe that if $u = x(x^2 + 3) = x^3 + 3x$, then $du = (3x^2 + 3) dx$ and so

$$\int \frac{x^2 + 1}{x(x^2 + 3)} dx = \frac{1}{3} \ln |x^3 + 3x| + C$$

Some nonrational functions can be changed into rational functions by means of appropriate substitutions. In particular, when an integrand contains an expression of the form $\sqrt[n]{g(x)}$, then the substitution $u = \sqrt[n]{g(x)}$ may be effective.

EXAMPLE 9 Evaluate $\int \frac{\sqrt{x+4}}{x} dx$.

SOLUTION Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = 2u du$. Therefore

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2 - 4} 2u du = 2 \int \frac{u^2}{u^2 - 4} du \\ &= 2 \int \left(1 + \frac{4}{u^2 - 4} \right) du\end{aligned}$$

We can evaluate this integral either by factoring $u^2 - 4$ as $(u - 2)(u + 2)$ and using partial fractions or by using Formula 6 with $a = 2$:

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= 2 \int du + 8 \int \frac{du}{u^2 - 4} \\ &= 2u + 8 \cdot \frac{1}{2 \cdot 2} \ln \left| \frac{u - 2}{u + 2} \right| + C \\ &= 2\sqrt{x+4} + 2 \ln \left| \frac{\sqrt{x+4} - 2}{\sqrt{x+4} + 2} \right| + C\end{aligned}$$

SOME SOLVED PROBLEMS

EXAMPLE 1 Evaluate $\int \frac{x^3 + 3x^2}{x^2 + 1} dx$.

Solution The numerator has degree 3 and the denominator has degree 2 so we need to divide. We use long division:

$$\begin{array}{r} x + 3 \\ x^2 + 1 \overline{) x^3 + 3x^2} \\ \underline{x^3 + x} \\ 3x^2 - x \\ \underline{3x^2 + 3} \\ -x - 3 \end{array}$$

$$\frac{x^3 + 3x^2}{x^2 + 1} = x + 3 - \frac{x + 3}{x^2 + 1}.$$

Thus,

$$\begin{aligned} \int \frac{x^3 + 3x^2}{x^2 + 1} dx &= \int (x + 3) dx - \int \frac{x}{x^2 + 1} dx - 3 \int \frac{dx}{x^2 + 1} \\ &= \frac{1}{2} x^2 + 3x - \frac{1}{2} \ln(x^2 + 1) - 3 \tan^{-1} x + C. \end{aligned}$$

EXAMPLE 2 Evaluate $\int \frac{x}{2x-1} dx$.

Solution The numerator and denominator have the same degree, 1, so division is again required. In this case the division can be carried out by manipulation of the integrand:

$$\frac{x}{2x-1} = \frac{1}{2} \frac{2x}{2x-1} = \frac{1}{2} \frac{2x-1+1}{2x-1} = \frac{1}{2} \left(1 + \frac{1}{2x-1} \right),$$

a process that we call *short division*

We have

$$\int \frac{x}{2x-1} dx = \frac{1}{2} \int \left(1 + \frac{1}{2x-1} \right) dx = \frac{x}{2} + \frac{1}{4} \ln |2x-1| + C.$$

LET US REMEMBER

The case of a linear denominator

$$\int \frac{1}{ax + b} dx = \frac{1}{a} \ln |ax + b| + C.$$

The case of a quadratic denominator

$$\int \frac{x dx}{x^2 + a^2} = \frac{1}{2} \ln(x^2 + a^2) + C,$$

$$\int \frac{x dx}{x^2 - a^2} = \frac{1}{2} \ln |x^2 - a^2| + C,$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C,$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C.$$

EXAMPLE 3Evaluate $\int \frac{(x+4)}{x^2-5x+6} dx$.**Solution** The partial fraction decomposition takes the form

$$\frac{x+4}{x^2-5x+6} = \frac{x+4}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}.$$

We calculate A and B by both of the methods suggested above.

Add the partial fractions

$$\frac{\quad}{x^2-5x+6} = \frac{Ax-3A+Bx-2B}{(x-2)(x-3)},$$

and equate the coefficient of x and the constant terms in the numerators on both sides to obtain

$$A+B=1 \quad \text{and} \quad -3A-2B=4.$$

Solve these equations to get $A=-6$ and $B=7$.

$$\begin{aligned} \int \frac{(x+4)}{x^2-5x+6} dx &= -6 \int \frac{1}{x-2} dx + 7 \int \frac{1}{x-3} dx \\ &= -6 \ln|x-2| + 7 \ln|x-3| + C. \end{aligned}$$

EXAMPLE 4 Evaluate $I = \int \frac{x^3 + 2}{x^3 - x} dx$.

Solution Since the numerator does not have degree smaller than the denominator, we must divide:

$$I = \int \frac{x^3 - x + x + 2}{x^3 - x} dx = \int \left(1 + \frac{x + 2}{x^3 - x} \right) dx = x + \int \frac{x + 2}{x^3 - x} dx.$$

Now we can use the method of partial fractions.

$$\begin{aligned} \frac{x + 2}{x^3 - x} &= \frac{x + 2}{x(x - 1)(x + 1)} = \frac{A}{x} + \frac{B}{x - 1} + \frac{C}{x + 1} \\ &= \frac{A(x^2 - 1) + B(x^2 + x) + C(x^2 - x)}{x(x - 1)(x + 1)} \end{aligned}$$

We have

$$\begin{array}{rclcl} A + B + C & = & 0 & \text{(coefficient of } x^2) \\ B - C & = & 1 & \text{(coefficient of } x) \\ -A & = & 2 & \text{(constant term).} \end{array}$$

It follows that $A = -2$, $B = 3/2$, and $C = 1/2$. We can also find these values using

Finally, we have

$$\begin{aligned} I &= x - 2 \int \frac{1}{x} dx + \frac{3}{2} \int \frac{1}{x - 1} dx + \frac{1}{2} \int \frac{1}{x + 1} dx \\ &= x - 2 \ln |x| + \frac{3}{2} \ln |x - 1| + \frac{1}{2} \ln |x + 1| + C. \end{aligned}$$

EXAMPLE 5 Evaluate $\int \frac{2 + 3x + x^2}{x(x^2 + 1)} dx$.

Solution Note that the numerator has degree 2 and the denominator degree 3, so no division is necessary. If we decompose the integrand as a sum of two simpler fractions, we want one with denominator x and one with denominator $x^2 + 1$. The appropriate form of the decomposition turns out to be

$$\frac{2 + 3x + x^2}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} = \frac{A(x^2 + 1) + Bx^2 + Cx}{x(x^2 + 1)}.$$

Note that corresponding to the quadratic (degree 2) denominator we use a linear (degree 1) numerator. Equating coefficients in the two numerators, we obtain

$$\begin{array}{rcll} A + B & = & 1 & \text{(coefficient of } x^2) \\ C & = & 3 & \text{(coefficient of } x) \\ A & = & 2 & \text{(constant term).} \end{array}$$

Hence $A = 2$, $B = -1$, and $C = 3$. We have, therefore,

$$\begin{aligned} \int \frac{2 + 3x + x^2}{x(x^2 + 1)} dx &= 2 \int \frac{1}{x} dx - \int \frac{x}{x^2 + 1} dx + 3 \int \frac{1}{x^2 + 1} dx \\ &= 2 \ln |x| - \frac{1}{2} \ln(x^2 + 1) + 3 \tan^{-1} x + C. \end{aligned}$$

Completing the Square

$$\begin{aligned}Ax^2 + Bx + C &= A \left(x^2 + \frac{B}{A}x + \frac{C}{A} \right) \\&= A \left(x^2 + \frac{B}{A}x + \frac{B^2}{4A^2} + \frac{C}{A} - \frac{B^2}{4A^2} \right) \\&= A \left(x + \frac{B}{2A} \right)^2 + \frac{4AC - B^2}{4A}.\end{aligned}$$

The substitution $u = x + \frac{B}{2A}$ should then be made.

EXAMPLE 6 Evaluate $I = \int \frac{1}{x^3 + 1} dx$.

Solution Here $Q(x) = x^3 + 1 = (x + 1)(x^2 - x + 1)$. The latter factor has no real roots, so it has no real linear subfactors. We have

$$\begin{aligned}\frac{1}{x^3 + 1} &= \frac{1}{(x + 1)(x^2 - x + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 - x + 1} \\ &= \frac{A(x^2 - x + 1) + B(x^2 + x) + C(x + 1)}{(x + 1)(x^2 - x + 1)}\end{aligned}$$

$$\begin{array}{rcll}A + B & & = 0 & \text{(coefficient of } x^2) \\ -A + B + C & = 0 & & \text{(coefficient of } x) \\ A & + C & = 1 & \text{(constant term).}\end{array}$$

Hence, $A = 1/3$, $B = -1/3$, and $C = 2/3$. We have

$$I = \frac{1}{3} \int \frac{dx}{x + 1} - \frac{1}{3} \int \frac{x - 2}{x^2 - x + 1} dx.$$

The first integral is easily evaluated; in the second we complete the square in the denominator: $x^2 - x + 1 = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$, and make a similar modification in the numerator.

$$\begin{aligned}
I &= \frac{1}{3} \ln |x + 1| - \frac{1}{3} \int \frac{x - \frac{1}{2} - \frac{3}{2}}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}} dx && \text{Let } u = x - 1/2, \\
&&& du = dx \\
&= \frac{1}{3} \ln |x + 1| - \frac{1}{3} \int \frac{u}{u^2 + \frac{3}{4}} du + \frac{1}{2} \int \frac{1}{u^2 + \frac{3}{4}} du \\
&= \frac{1}{3} \ln |x + 1| - \frac{1}{6} \ln \left(u^2 + \frac{3}{4}\right) + \frac{1}{2} \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2u}{\sqrt{3}}\right) + C \\
&= \frac{1}{3} \ln |x + 1| - \frac{1}{6} \ln(x^2 - x + 1) + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x - 1}{\sqrt{3}}\right) + C.
\end{aligned}$$

EXAMPLE 7Evaluate $\int \frac{1}{x(x-1)^2} dx$.**Solution** The appropriate partial fraction decomposition here is

$$\begin{aligned}\frac{1}{x(x-1)^2} &= \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2} \\ &= \frac{A(x^2 - 2x + 1) + B(x^2 - x) + Cx}{x(x-1)^2}.\end{aligned}$$

Equating coefficients of x^2 , x , and 1 in the numerators of both sides, we get

$$\begin{array}{rcll}A + B & = & 0 & \text{(coefficient of } x^2\text{)} \\ -2A - B + C & = & 0 & \text{(coefficient of } x\text{)} \\ A & = & 1 & \text{(constant term).}\end{array}$$

Hence $A = 1$, $B = -1$, $C = 1$, and

$$\begin{aligned}\int \frac{1}{x(x-1)^2} dx &= \int \frac{1}{x} dx - \int \frac{1}{x-1} dx + \int \frac{1}{(x-1)^2} dx \\ &= \ln|x| - \ln|x-1| - \frac{1}{x-1} + C \\ &= \ln\left|\frac{x}{x-1}\right| - \frac{1}{x-1} + C.\end{aligned}$$

EXAMPLE 8Evaluate $I = \int \frac{x^2 + 2}{4x^5 + 4x^3 + x} dx$.

Solution The denominator factors to $x(2x^2 + 1)^2$, so the appropriate partial fraction decomposition is

$$\begin{aligned}\frac{x^2 + 2}{x(2x^2 + 1)^2} &= \frac{A}{x} + \frac{Bx + C}{2x^2 + 1} + \frac{Dx + E}{(2x^2 + 1)^2} \\ &= \frac{A(4x^4 + 4x^2 + 1) + B(2x^4 + x^2) + C(2x^3 + x) + Dx^2 + Ex}{x(2x^2 + 1)^2}.\end{aligned}$$

Thus

$$\begin{array}{rcll}4A & + & 2B & = 0 & \text{(coefficient of } x^4\text{)} \\ & & 2C & = 0 & \text{(coefficient of } x^3\text{)} \\ 4A & + & B & + & D & = 1 & \text{(coefficient of } x^2\text{)} \\ & & C & + & E & = 0 & \text{(coefficient of } x\text{)} \\ A & & & & & = 2 & \text{(constant term).}\end{array}$$

Solving these equations, we get $A = 2$, $B = -4$, $C = 0$, $D = -3$, and $E = 0$.

$$\begin{aligned}I &= 2 \int \frac{dx}{x} - 4 \int \frac{x dx}{2x^2 + 1} - 3 \int \frac{x dx}{(2x^2 + 1)^2} && \text{Let } u = 2x^2 + 1, \\ &&& du = 4x dx \\ &= 2 \ln |x| - \int \frac{du}{u} - \frac{3}{4} \int \frac{du}{u^2} \\ &= 2 \ln |x| - \ln |u| + \frac{3}{4u} + C \\ &= \ln \left(\frac{x^2}{2x^2 + 1} \right) + \frac{3}{4} \frac{1}{2x^2 + 1} + C.\end{aligned}$$

$$\int \frac{1}{1 + \sqrt{2x}} dx$$

$$\begin{aligned}\text{Let } 2x &= u^2, \\ 2 dx &= 2u du\end{aligned}$$

$$= \int \frac{u}{1 + u} du$$

$$= \int \frac{1 + u - 1}{1 + u} du$$

$$= \int \left(1 - \frac{1}{1 + u} \right) du$$

$$\begin{aligned}\text{Let } v &= 1 + u, \\ dv &= du\end{aligned}$$

$$= u - \int \frac{dv}{v} = u - \ln |v| + C$$

$$= \sqrt{2x} - \ln(1 + \sqrt{2x}) + C$$

REMEMBER

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

$$\begin{aligned}\int \sec x \, dx &= \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} \, dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx\end{aligned}$$

If we substitute $u = \sec x + \tan x$, then $du = (\sec x \tan x + \sec^2 x) \, dx$, so the integral becomes $\int (1/u) \, du = \ln |u| + C$. Thus, we have

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

EXAMPLE 1 Use partial fractions to evaluate

$$\int \frac{x^2 + 4x + 1}{(x - 1)(x + 1)(x + 3)} dx.$$

Solution The partial fraction decomposition has the form

$$\frac{x^2 + 4x + 1}{(x - 1)(x + 1)(x + 3)} = \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{C}{x + 3}.$$

To find the values of the undetermined coefficients A , B , and C , we clear fractions and get

$$\begin{aligned} x^2 + 4x + 1 &= A(x + 1)(x + 3) + B(x - 1)(x + 3) + C(x - 1)(x + 1) \\ &= A(x^2 + 4x + 3) + B(x^2 + 2x - 3) + C(x^2 - 1) \\ &= (A + B + C)x^2 + (4A + 2B)x + (3A - 3B - C). \end{aligned}$$

The polynomials on both sides of the above equation are identical, so we equate coefficients of like powers of x , obtaining

$$\begin{array}{ll} \text{Coefficient of } x^2: & A + B + C = 1 \\ \text{Coefficient of } x^1: & 4A + 2B = 4 \\ \text{Coefficient of } x^0: & 3A - 3B - C = 1 \end{array}$$

There are several ways of solving such a system of linear equations for the unknowns A , B , and C , including elimination of variables or the use of a calculator or computer. Whatever method is used, the solution is $A = 3/4$, $B = 1/2$, and $C = -1/4$. Hence we have

$$\begin{aligned} \int \frac{x^2 + 4x + 1}{(x - 1)(x + 1)(x + 3)} dx &= \int \left[\frac{3}{4} \frac{1}{x - 1} + \frac{1}{2} \frac{1}{x + 1} - \frac{1}{4} \frac{1}{x + 3} \right] dx \\ &= \frac{3}{4} \ln |x - 1| + \frac{1}{2} \ln |x + 1| - \frac{1}{4} \ln |x + 3| + K, \end{aligned}$$

where K is the arbitrary constant of integration (to avoid confusion with the undetermined coefficient we labeled as C). ■

EXAMPLE 2 Use partial fractions to evaluate

$$\int \frac{6x + 7}{(x + 2)^2} dx.$$

Solution First we express the integrand as a sum of partial fractions with undetermined coefficients.

$$\frac{6x + 7}{(x + 2)^2} = \frac{A}{x + 2} + \frac{B}{(x + 2)^2}$$

$$6x + 7 = A(x + 2) + B \quad \text{Multiply both sides by } (x + 2)^2.$$

$$= Ax + (2A + B)$$

Equating coefficients of corresponding powers of x gives

$$A = 6 \quad \text{and} \quad 2A + B = 7, \quad \text{or} \quad A = 6 \quad \text{and} \quad B = -5.$$

Therefore,

$$\begin{aligned} \int \frac{6x + 7}{(x + 2)^2} dx &= \int \left(\frac{6}{x + 2} - \frac{5}{(x + 2)^2} \right) dx \\ &= 6 \int \frac{dx}{x + 2} - 5 \int (x + 2)^{-2} dx \\ &= 6 \ln |x + 2| + 5(x + 2)^{-1} + C. \end{aligned}$$

EXAMPLE 3 Use partial fractions to evaluate

$$\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} dx.$$

Solution First we divide the denominator into the numerator to get a polynomial plus a proper fraction.

$$\begin{array}{r} 2x \\ x^2 - 2x - 3 \overline{) 2x^3 - 4x^2 - x - 3} \\ \underline{2x^3 - 4x^2 - 6x} \\ 5x - 3 \end{array}$$

Then we write the improper fraction as a polynomial plus a proper fraction.

$$\frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} = 2x + \frac{5x - 3}{x^2 - 2x - 3}$$

We found the partial fraction decomposition of the fraction on the right in the opening example, so

$$\begin{aligned} \int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3} dx &= \int 2x \, dx + \int \frac{5x - 3}{x^2 - 2x - 3} dx \\ &= \int 2x \, dx + \int \frac{2}{x + 1} dx + \int \frac{3}{x - 3} dx \\ &= x^2 + 2 \ln |x + 1| + 3 \ln |x - 3| + C. \end{aligned}$$



EXAMPLE 4 Use partial fractions to evaluate

$$\int \frac{-2x + 4}{(x^2 + 1)(x - 1)^2} dx.$$

Solution The denominator has an irreducible quadratic factor as well as a repeated linear factor, so we write

$$\frac{-2x + 4}{(x^2 + 1)(x - 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x - 1} + \frac{D}{(x - 1)^2}. \quad (2)$$

Clearing the equation of fractions gives

$$\begin{aligned}-2x + 4 &= (Ax + B)(x - 1)^2 + C(x - 1)(x^2 + 1) + D(x^2 + 1) \\ &= (A + C)x^3 + (-2A + B - C + D)x^2 \\ &\quad + (A - 2B + C)x + (B - C + D).\end{aligned}$$

Equating coefficients of like terms gives

$$\begin{array}{ll}\text{Coefficients of } x^3: & 0 = A + C \\ \text{Coefficients of } x^2: & 0 = -2A + B - C + D \\ \text{Coefficients of } x^1: & -2 = A - 2B + C \\ \text{Coefficients of } x^0: & 4 = B - C + D\end{array}$$

We solve these equations simultaneously to find the values of A , B , C , and D :

$$\begin{array}{ll} -4 = -2A, & A = 2 & \text{Subtract fourth equation from second.} \\ C = -A = -2 & & \text{From the first equation} \\ B = (A + C + 2)/2 = 1 & & \text{From the third equation and } C = -A \\ D = 4 - B + C = 1. & & \text{From the fourth equation} \end{array}$$

We substitute these values into Equation (2), obtaining

$$\frac{-2x + 4}{(x^2 + 1)(x - 1)^2} = \frac{2x + 1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2}.$$

Finally, using the expansion above we can integrate:

$$\begin{aligned}\int \frac{-2x + 4}{(x^2 + 1)(x - 1)^2} dx &= \int \left(\frac{2x + 1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2} \right) dx \\ &= \int \left(\frac{2x}{x^2 + 1} + \frac{1}{x^2 + 1} - \frac{2}{x - 1} + \frac{1}{(x - 1)^2} \right) dx \\ &= \ln(x^2 + 1) + \tan^{-1} x - 2 \ln |x - 1| - \frac{1}{x - 1} + C. \quad \blacksquare\end{aligned}$$

EXAMPLE 5 Use partial fractions to evaluate

$$\int \frac{dx}{x(x^2 + 1)^2}.$$

Solution The form of the partial fraction decomposition is

$$\frac{1}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}.$$

Multiplying by $x(x^2 + 1)^2$, we have

$$\begin{aligned} 1 &= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x \\ &= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex \\ &= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A \end{aligned}$$

If we equate coefficients, we get the system

$$A + B = 0, \quad C = 0, \quad 2A + B + D = 0, \quad C + E = 0, \quad A = 1.$$

Solving this system gives $A = 1$, $B = -1$, $C = 0$, $D = -1$, and $E = 0$. Thus,

$$\begin{aligned} \int \frac{dx}{x(x^2 + 1)^2} &= \int \left[\frac{1}{x} + \frac{-x}{x^2 + 1} + \frac{-x}{(x^2 + 1)^2} \right] dx \\ &= \int \frac{dx}{x} - \int \frac{x dx}{x^2 + 1} - \int \frac{x dx}{(x^2 + 1)^2} \\ &= \int \frac{dx}{x} - \frac{1}{2} \int \frac{du}{u} - \frac{1}{2} \int \frac{du}{u^2} && \begin{array}{l} u = x^2 + 1, \\ du = 2x dx \end{array} \\ &= \ln |x| - \frac{1}{2} \ln |u| + \frac{1}{2u} + K \\ &= \ln |x| - \frac{1}{2} \ln (x^2 + 1) + \frac{1}{2(x^2 + 1)} + K \\ &= \ln \frac{|x|}{\sqrt{x^2 + 1}} + \frac{1}{2(x^2 + 1)} + K. \end{aligned}$$

